

State Transition

- New: just been created, not ready for the queue
- Ready: can be loaded by the OS
- Running: scheduler has picked this process from the queue and executed it, usually only one
- Blocked: not in the queue, waiting
- Exit: finished, needs to terminate

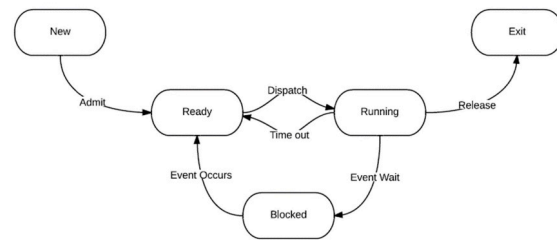


Figure T3_1

- Admit signal: process is fully loaded into memory and control is established
- Dispatch signal: scheduler assigns CPU to the process
- Time out signal: expired or preempted, pushed back to the queue
- Event Wait/Event Occurs: generally requests that cannot be met at the moment, has to wait until something occurs
 - OS not ready for a service
 - Unavailable resource
 - Wait for an input
- Release Signal: release resources and end the process

Ex: How is the state transition triggered?

- OS scheduling,
- external interrupts
- program requests

A process can have

- A CPU time allocation (virtual CPU, the role of OS)
- Memory (real or virtual)
- Process ID
- Threads

The Process Control Block (PCB) or Task Control Block (TCB) maintains all the relevant information for the process. a kind of struct in C. Including:

- Process ID
- Process state
- PC, SP and other registers (stored)
- Scheduling information (priority)
- User information
- Other information

Program vs process vs thread:

- Program usually refers to the instructions that are stored on **disk**
- Process is the program with execution **context**
- Some OSs may use the term “task”, particularly in an embedded system context.

We will use both terms interchangeably for this course

- Thread: a lightweight process; a process may have multiple threads which **share the same system resources** - faster creation, termination, switching and communication